

Andrew Lepetri

Software Engineer



lepetriandy@gmail.com



619-971-0042



linkedin.com/in/andrew-lepetri



github.com/alepetri



alepetri.github.io

Professional Experience

Senior Autonomy Platform Software Engineer

Joby Aviation | June 2024 - Current | Santa Cruz, CA (Remote Employee)

- Authored flight computer drivers to communicate with avionics sensors, actuators, and transponders
- Fortified and expanded foundational flight code framework
- Improved and maintained a Bazel monorepo, limiting dependency tree size, creating custom rules and aspects, linking flight code, simulation, analysis, and ground control software
- Integrated a Systems Engineering tool/ database with our software stack leveraging Rest API hooks, linking software unit and integration tests to automatically report status to the tool

Senior Software Engineer

Xwing (Acquired by Joby Aviation) | April 2022 - May 2024 | San Francisco, CA (Remote Employee)

- Simulated avionics boxes and their hardware interface protocols to provide varying levels of integration test capabilities, from Software, to Hardware, to Aircraft-in-the-loop
- Performed trade studies on hardware acquisition for a Hardware-In-the-Loop (HIL) lab
- Authored a comprehensive ARINC 429 driver utility library in C
- Developed and maintained a C++ XPlane12 plugin that positioned the camera, injected objects, and calculated bounding boxes for a rendered scenario generating datasets for landing and takeoff vision based Machine Learning models
- Helped craft a middleware framework to sit between an ARINC 653 compliant OS and user applications that could be assigned to a partition, scheduled, managed, and perform Inter/Intra Partition Communication
- Designed a web application that allowed engineers to pull, visualize, edit, and push updates to a requirements management tool. Written in Typescript, leveraging PNPM for package management, Next.JS as a front and backend framework, a

Skills

Programming Languages

C++, Python, Rust, C, TypeScript, MATLAB, C#

Technical Tools

Bazel, Cmake, Git, Ubuntu, Vscode, Next.js, Django, Docker, PostgreSQL

Education

Master of Science in Computer Science

Georgia Institute of Technology | 2023 | Atlanta, GA | 3.7 GPA

- Specialization in Computational Perception and Robotics

Bachelor of Science in Mechanical Engineering with a Minor in Computer Science

San Diego State University | 2018 | San Diego, CA | 3.54 GPA

- Awarded Dean's List Honors (2014-2018)

PostgreSQL instance for persistent storage, and prisma for the database Object Relational Mapping (ORM)

- Bridged Systems and Software by conforming and linking code to requirements and test cases
- Maintained consistency with Simulink and C flight code through bi-directional automated exports and checks

Systems Engineer

Northrop Grumman | March 2019 - March 2022 | Rancho Bernardo, CA

- Conducted Modeling, Simulation, and Analysis in support of future capabilities of an Unmanned Aircraft System (UAS)
- Performed Risk Reduction by modeling Sense-And-Avoid functionality onto a UAS, flying the emulated UAS in various airspace environments to collect and analyze critical safety metrics, characterizing the benefits and challenges potential additions would have on the system
- Practiced requirement generation from industry standards
- Integrated physical LRUs into a Hardware in the Loop (HIL) simulation
- Created Command and Control Graphical User Interfaces in C#
- Emulated hardware interfaces in C++: encoding, piping, and handling messages between subsystems
- Developed a Data Analysis framework in Python using Django, Postgres, and Docker with real-time simulation monitoring capabilities
- Executed trade studies using Monte Carlo runs to generate statistics to aid decision making
- Worked with an Agile team of engineers to develop and improve software using Git configuration control
- Created process automation tools in Python to increase operational efficiency

Vision Engineer (Internship)

Cohu Inc | March 2017 - May 2018 | Poway, CA

- Contributed within a cross-disciplinary team to research, design, develop, test, and present vision related solutions in processing and sorting thousands of microchips per hour
- Worked on an array of projects including camera and lighting assemblies, micron precision alignment, device shuttle trays, OCR and 2DID recognition, and vision software applications

Projects

NGC Von Kármán Competition (2020)

- Performed 3D target tracking using sensor fusion of camera and thermal data to update positional belief within an Extended Kalman Filter.

SDSU Mechatronics (2015-2018)

- Project manager for a team of Mechanical Engineers researching and designing an Autonomous Underwater Vehicle to compete in the AUVSI Robo Sub competition.
- Developed leadership and interpersonal skills through a simulated work environment.

SDSU IEEE's Aztec Air (2016-2017)

- Leader of a team of Electrical and Computer Engineers in constructing and programming a Semi-Autonomous Aerial Vehicle that utilizes inflight sensory interpolation to adjust flight path.

Instructional Assistant

San Diego State University | August 2016 - December 2017 | San Diego, CA

- Taught and tutored for SDSU's MATLAB Computer Applications course
- Developed mathematics based assignments to be solved in MATLAB
- Honed teaching abilities through delivering weekly recitations sessions to students
- Unpacked complex engineering and programming topics into understandable lessons

Biomechanical Researcher

San Diego State University | January 2016 - April 2016 | San Diego, CA

- Performed fiber-based cell migration simulations in MATLAB
- Modeled key mechanical properties of cells, analyzing their chosen paths through a randomly generated Extracellular Matrix
- Sourced simulation parameters from scientific journals

Interests



Birding, Hiking, Traveling



Crocheting, Knitting



3D Printing, Tinkering, Laser Cutting



Board Games, Chess, Game Dev